

Ekansh Chauhan

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Summary

Results-driven software developer with hands-on experience in full-stack software development and AI systems. Proficient in designing and delivering scalable REST API backends and production-grade RAG-based applications. Adept at end-to-end software development lifecycle from system design and implementation to testing and optimization. Strong foundation in data structures, algorithms, and object-oriented programming with a track record of measurable performance improvements across academic and independent software development projects.

Education

The LNM Institute of Information Technology

Bachelor of Technology in Computer Science and Engineering — CGPA: 9.36/10

Jaipur, India

July 2023 – June 2027

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Computer Networks, Database Management Systems, Analysis of Algorithms

Technical Skills

- **Languages:** C, C++, Python, JavaScript, SQL
- **Frameworks & Libraries:** React.js, Node.js, Express.js, FastAPI, LangChain, LangGraph, Redux Toolkit, Tailwind CSS
- **Databases:** MySQL, MongoDB, ChromaDB, SQLite, FAISS
- **Tools & Platforms:** Git, GitHub, REST APIs, JWT, Postman, Ollama, Stripe
- **Concepts:** Data Structures and Algorithms, System Design, Retrieval-Augmented Generation (RAG), Agentic AI

Projects

Adaptive Tutoring System (RAG-Based) *Python, FastAPI, LangChain, FAISS, Groq (Llama-3), HuggingFace, SQLite, Ragas*

- Engineered a hybrid RAG pipeline combining FAISS dense retrieval and TF-IDF sparse retrieval, achieving **83% retrieval precision** with latency below **12 ms/query**.
- Integrated **Reciprocal Rank Fusion (RRF)** for multi-source aggregation and a **Corrective RAG (CRAG)** module with LLM-based relevance filtering, cutting context payload by **56%**.
- Designed an adaptive learning engine personalizing explanation depth based on student performance history, validated via a custom **Ragas** evaluation framework using Groq LLM-as-a-Judge.

Trip Sage — Intelligent Travel Planner

React.js, Node.js, Express.js, MongoDB, JWT, Python

- Delivered a full-stack **MERN** application for automated multi-day itinerary generation, integrated with Google Places, Traffic, and Weather REST APIs with periodic live data refresh.
- Developed a route optimization module using **K-Means Clustering** and a custom **Proximity Sorting** algorithm, achieving a **54.6% reduction** in total travel distance versus baseline benchmarks.
- Secured all API endpoints with JWT-based authentication and role-based access control across the full software development lifecycle from design to deployment.

Autonomous Closed-Loop Fault Localization and Program Repair Agent *Python, ChromaDB, AST, Groq, Subprocess*

- Architected a multi-stage agentic debugging pipeline with chained LLM roles (Architect, Reviewer, Test Engineer) to autonomously localize and patch bugs from runtime crash logs, reducing token consumption by **70%+** via AST-driven static code slicing.
- Constructed a local RAG system using ChromaDB to semantically index codebase documentation, dynamically surfacing top-3 relevant classes from stack traces instead of transmitting the full codebase.
- Shipped a closed-loop self-correction engine validating LLM-generated patches through a two-stage sandbox (syntax check then subprocess execution), auto-retrying up to 3 times on failure.

Honors and Achievements

- **LeetCode Knight** — Attained Knight rank placing in the top tier of competitive programmers globally; solved problems across dynamic programming, graphs, and system design.
- **Merit Scholarship** — LNMIIT Foundation Day Award for academic excellence in Year 1 and Year 2; ranked **3rd** in the Y23 batch of the Computer Science program.

Hackathons

- **3rd Place** at Hackstorm 2025, JEC Kukas — competed against **80+** teams, delivering a production-ready prototype under a 24-hour constraint.
- **Top 40 of 1200+ Teams** at Hackground India 2025 — contributed core backend architecture including API design, database schema, and server-side business logic.